

Measures

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Remark

Remark. A measure assigns a size to each set in a collection of sets: how many elements it has, how long it is, or how likely it is. This page builds measures from σ -algebras and additivity, using the [sets and functions](#) page's definitions. The Morrison Lab's probability notes define a probability measure as a measure that gives the whole space size 1 ([probability measure](#)).

σ -algebras

Definition 0.1 (σ -algebra). A σ -algebra on a set S is a collection $\mathcal{S}$ of [subsets](#) of S that satisfies:

- $\mathcal{S}$ contains S itself.
- For each set A in $\mathcal{S}$, $\mathcal{S}$ contains its complement $S \setminus A$.
- For each sequence $A_1, A_2, \dots$ of sets in $\mathcal{S}$, $\mathcal{S}$ contains their union $\bigcup_{i=1}^{\infty} A_i$.

Remark

Remark. Other sources call it a σ -field (see [Wikipedia: \$\sigma\$ -algebra](#)).

Example 0.1 (σ -algebras for die rolls). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ ([sets of die rolls](#)), the collection of all subsets of D is a σ -algebra on D , since complements and unions of subsets of D are again subsets of D . So is the smaller collection

$$\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$$

which contains D , the complement of each of its sets, and every union of its sets: for

example, $\{2, 4, 6\} \cup \{1, 3, 5\} = D$.

Theorem 0.1 (Closure properties of a σ -algebra). *If $\mathcal{S}$ is a σ -algebra on a set S , then $\mathcal{S}$ contains:*

- the empty set $\emptyset$;
- the union $A_1 \cup \dots \cup A_n$ of any finitely many sets $A_1, \dots, A_n$ in $\mathcal{S}$;
- the intersection $\bigcap_{i=1}^{\infty} A_i$ of any sequence $A_1, A_2, \dots$ of sets in $\mathcal{S}$.

i Proof

Proof. Empty set. $\mathcal{S}$ contains S , so it contains the complement of S , which is $S \setminus S = \emptyset$.
Finite unions. Extend $A_1, \dots, A_n$ to a sequence by setting $A_{n+1} = A_{n+2} = \dots = \emptyset$; each of these sets is in $\mathcal{S}$, by the first part. Then:

$$\begin{aligned} A_1 \cup \dots \cup A_n &= A_1 \cup \dots \cup A_n \cup \emptyset \cup \emptyset \cup \dots \quad (\text{a union with } \emptyset \text{ adds no elements}) \\ &= \bigcup_{i=1}^{\infty} A_i \quad (A_i = \emptyset \text{ for } i > n) \end{aligned}$$

and $\mathcal{S}$ contains $\bigcup_{i=1}^{\infty} A_i$ by the union rule of Definition 0.1.

Countable intersections. An element of S is in every A_i exactly when it is in none of the complements $S \setminus A_i$, so:

$$\bigcap_{i=1}^{\infty} A_i = S \setminus \bigcup_{i=1}^{\infty} (S \setminus A_i) \quad (\text{De Morgan's law})$$

Each $S \setminus A_i$ is in $\mathcal{S}$ by the complement rule, so their union is in $\mathcal{S}$ by the union rule, and the complement of that union is in $\mathcal{S}$ by the complement rule again. $\square$

Theorem 0.2 (All subsets form a σ -algebra). *For any set S , the collection of all subsets of S is a σ -algebra on S .*

i Proof

Proof. Each condition of Definition 0.1 holds:

- S is a subset of itself.
- For each subset A of S , the complement $S \setminus A$ contains only elements of S , so it is a subset of S .
- For each sequence $A_1, A_2, \dots$ of subsets of S , every element of $\bigcup_{i=1}^{\infty} A_i$ is in some A_i and so in S ; the union is a subset of S .

□

Pairwise disjoint sets

Definition 0.2 (Pairwise disjoint sets). Finitely or countably many sets $A_1, A_2, \dots$ are **pairwise disjoint** (also called **disjoint** or **mutually disjoint**) when no two of them share an element:

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

i Remark

Remark. In probability, pairwise disjoint events are usually called **mutually exclusive**.

Example 0.2 (Low and high die rolls). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (**sets of die rolls**), the sets $\{1, 2\}$ and $\{5, 6\}$ are pairwise disjoint: no roll is in both. The sets $\{2, 4, 6\}$ and $\{1, 2\}$ are not, because 2 is in both.

Additivity

Definition 0.3 (Finite additivity). Let $\mathcal{S}$ be a **σ -algebra** on a set S . A **function** $\mu : \mathcal{S} \rightarrow [0, \infty]$, with values in the **extended non-negative reals**, is **finitely additive** if, for every finite collection of **pairwise disjoint** sets $A_1, \dots, A_n$ in $\mathcal{S}$, the value of their union is the sum of their values:

$$\mu(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mu(A_i)$$

Example 0.3 (Counting elements is finitely additive). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (**sets of die rolls**), let $\mu(A) \stackrel{\text{def}}{=} |A|$, the number of elements of A , for each set A in the σ -algebra of all subsets of D (Example 0.1). The sets $\{1, 2\}$ and $\{5, 6\}$ are pairwise disjoint (Example 0.2), and:

$$\begin{aligned}
\mu(\{1, 2\} \cup \{5, 6\}) &= \mu(\{1, 2, 5, 6\}) && \text{(take the union)} \\
&= 4 && \text{(count the elements)} \\
&= 2 + 2 && \text{(write 4 as a sum)} \\
&= \mu(\{1, 2\}) + \mu(\{5, 6\}) && \text{(count each set's elements)}
\end{aligned}$$

The same holds for any pairwise disjoint sets, because the sizes of disjoint sets add, so μ is finitely additive.

Definition 0.4 (Countable additivity). Let $\mathcal{S}$ be a σ -algebra on a set S . A function $\mu : \mathcal{S} \rightarrow [0, \infty]$ is **countably additive** (also called **σ -additive**) if, for every sequence of pairwise disjoint sets $A_1, A_2, \dots$ in $\mathcal{S}$, the value of their union is the sum of their values:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Example 0.4 (Counting elements is countably additive). For the counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.3, take any sequence of pairwise disjoint sets $A_1, A_2, \dots$. No two of them share an element, and D has only six elements, so at most six of the A_i contain any elements; the rest are $\emptyset$, with $\mu(\emptyset) = 0$. The infinite sum therefore has at most six nonzero terms, and finite additivity (Example 0.3) shows that those terms add up to μ of the union. So μ is countably additive.

Lemma 0.1 (Sums of non-negative terms). Let $a_1, a_2, \dots$ be values in $[0, \infty]$, with partial sums $s_n \stackrel{\text{def}}{=} \sum_{i=1}^n a_i$, where $x + \infty = \infty$ for every x in $[0, \infty]$. Then the partial sums are non-decreasing, $s_1 \leq s_2 \leq \dots$, so their limit, the sum $\sum_{i=1}^{\infty} a_i = \lim_{n \rightarrow \infty} s_n$, always exists, and it is either a finite number or ∞ .

Proof

Proof. For each n :

$$\begin{aligned}
s_{n+1} &= s_n + a_{n+1} && \text{(definition of } s_{n+1}\text{)} \\
&\geq s_n && (a_{n+1} \geq 0)
\end{aligned}$$

If some partial sum s_N is ∞ , then $s_n = \infty$ for every $n \geq N$, so the limit is ∞ . Otherwise, the partial sums form a non-decreasing sequence of real numbers. If that sequence is bounded above, it converges to a finite number (see [Wikipedia: Monotone convergence theorem](#)). If it is not bounded above, then for every number M some s_N exceeds M , and so does every later $s_n \geq s_N$; the limit is ∞ . $\square$

i Remark

Remark. By Lemma 0.1, the right-hand side of the equation in Definition 0.4 always has a value.

Lemma 0.2 (Countable additivity and the empty set). *If μ is a countably additive function on a σ -algebra $\mathcal{S}$, then $\mu(\emptyset) = 0$ or $\mu(\emptyset) = \infty$.*

i Proof

Proof. The set $\emptyset = S \setminus S$ is in $\mathcal{S}$, as the complement of S , and the sequence $\emptyset, \emptyset, \dots$ is pairwise disjoint with union $\emptyset$, so:

$$\begin{aligned}\mu(\emptyset) &= \mu\left(\bigcup_{i=1}^{\infty} \emptyset\right) \quad (\text{the union of copies of } \emptyset \text{ is } \emptyset) \\ &= \sum_{i=1}^{\infty} \mu(\emptyset) \quad (\text{countable additivity})\end{aligned}$$

If $\mu(\emptyset) = c$ for a finite $c > 0$, the right-hand side is $c + c + \dots = \infty \neq c$, a contradiction. So $\mu(\emptyset)$ is 0 or ∞ . $\square$

Example 0.5 (A countably additive function with $\mu(\emptyset) = 0$). The counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.4 is countably additive, and:

$$\begin{aligned}\mu(\emptyset) &= |\emptyset| \quad (\text{definition of } \mu) \\ &= 0 \quad (\emptyset \text{ has no elements})\end{aligned}$$

Example 0.6 (A countably additive function with $\mu(\emptyset) = \infty$). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (sets of die rolls), let $\mu(A) \stackrel{\text{def}}{=} \infty$ for every set A in the σ -algebra of all subsets of D (Example 0.1), including $A = \emptyset$. For any sequence of pairwise disjoint sets $A_1, A_2, \dots$, the left-hand side of the countable additivity equation is:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \infty \quad (\text{definition of } \mu)$$

and the right-hand side, the limit of its partial sums (Lemma 0.1), is:

$$\begin{aligned}\sum_{i=1}^{\infty} \mu(A_i) &= \infty + \infty + \dots \quad (\text{definition of } \mu) \\ &= \infty \quad (\text{every partial sum is } \infty)\end{aligned}$$

The two sides agree, so μ is countably additive, and $\mu(\emptyset) = \infty$.

i Remark

Remark. Example 0.5 and Example 0.6 show that both values allowed by Lemma 0.2 occur. So $\mu(\emptyset) = 0$ is an extra requirement, not a consequence of countable additivity.

Theorem 0.3 (Countable additivity implies finite additivity). *If μ is a countably additive function on a σ -algebra $\mathcal{S}$, and $\mu(\emptyset) = 0$, then μ is finitely additive.*

i Proof

Proof. Let $A_1, \dots, A_n$ be pairwise disjoint sets in $\mathcal{S}$, and extend them to a sequence by setting $A_{n+1} = A_{n+2} = \dots = \emptyset$. The extended sequence is still pairwise disjoint, since $\emptyset$ shares no element with any set, and its union is $A_1 \cup \dots \cup A_n$. So:

$$\begin{aligned} \mu(A_1 \cup \dots \cup A_n) &= \mu\left(\bigcup_{i=1}^{\infty} A_i\right) && (A_i = \emptyset \text{ for } i > n) \\ &= \sum_{i=1}^{\infty} \mu(A_i) && (\text{countable additivity}) \\ &= \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{\infty} \mu(\emptyset) && (A_i = \emptyset \text{ for } i > n) \\ &= \sum_{i=1}^n \mu(A_i) && (\mu(\emptyset) = 0) \end{aligned}$$

□

Example 0.7 (Finitely additive but not countably additive). Let $S = \{0, 1, 2, \dots\}$, with the σ -algebra of all subsets of S (Theorem 0.2), and define:

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

μ is finitely additive. Let $A_1, \dots, A_n$ be pairwise disjoint subsets of S . If every A_i is finite, then so is their union, and:

$$\begin{aligned}
\mu(A_1 \cup \dots \cup A_n) &= 0 && \text{(a finite union of finite sets is finite)} \\
&= \sum_{i=1}^n 0 && \text{(a sum of zeros is 0)} \\
&= \sum_{i=1}^n \mu(A_i) && \text{(each } A_i \text{ is finite)}
\end{aligned}$$

If some A_k is infinite, then the union, which contains A_k , is infinite too, and:

$$\begin{aligned}
\mu(A_1 \cup \dots \cup A_n) &= \infty && \text{(the union is infinite)} \\
&= \mu(A_k) + \sum_{i \neq k} \mu(A_i) && (\mu(A_k) = \infty, \text{ and } \infty + x = \infty) \\
&= \sum_{i=1}^n \mu(A_i) && \text{(regroup the terms)}
\end{aligned}$$

μ is not countably additive. The single-element sets $\{0\}, \{1\}, \{2\}, \dots$ are pairwise disjoint, and their union is S , which is infinite. So:

$$\begin{aligned}
\mu\left(\bigcup_{k=0}^{\infty} \{k\}\right) &= \mu(S) && \text{(the union is } S) \\
&= \infty && (S \text{ is infinite})
\end{aligned}$$

but:

$$\begin{aligned}
\sum_{k=0}^{\infty} \mu(\{k\}) &= 0 + 0 + \dots && \text{(each } \{k\} \text{ is finite)} \\
&= 0 && \text{(every partial sum is 0)}
\end{aligned}$$

i Remark

Remark. Example 0.7 shows that the converse of Theorem 0.3 fails, even when $\mu(\emptyset) = 0$: a finitely additive function need not be countably additive (see [Wikipedia: Sigma-additive set function — An additive function which is not \$\sigma\$ -additive](#)).

Measures

Definition 0.5 (Measure). A **measure** on a set S with a σ -algebra $\mathcal{S}$ is a function $\mu : \mathcal{S} \rightarrow [0, \infty]$ that satisfies:

- $\mu(\emptyset) = 0$.
- μ is **countably additive**.

Example 0.8 (Counting elements is a measure). The counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.3, defined on the σ -algebra of all subsets of D (Example 0.1), takes values in $[0, \infty]$, is countably additive (Example 0.4), and gives $\mu(\emptyset) = 0$, so it is a measure.

Corollary 0.1 (Measures are finitely additive). *Every measure is finitely additive.*

i Proof

Proof. A measure μ is countably additive and has $\mu(\emptyset) = 0$ (Definition 0.5), so Theorem 0.3 applies to it. $\square$

Definition 0.6 (Counting measure). The **counting measure** on a set S is the **measure** on the σ -algebra of all subsets of S that assigns each finite set its number of elements, and each infinite set the value ∞ :

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

Example 0.9 (Counting measure on the non-negative integers). For the counting measure μ on $\{0, 1, 2, \dots\}$, $\mu(\{0, 1, 2\}) = 3$, and the set of even numbers has $\mu(\{0, 2, 4, \dots\}) = \infty$. The even numbers are the union of the pairwise disjoint sets $\{0\}, \{2\}, \{4\}, \dots$, and countable additivity agrees: $\sum_{k=0}^{\infty} \mu(\{2k\}) = 1 + 1 + \dots = \infty$.

i Remark

Remark. A **measure** generalizes size: it can measure how many elements a set has, as in Definition 0.6, or how long a set of real numbers is, as **Lebesgue measure** does, assigning each interval $[a, b]$ its length $b - a$. Both appear as reference measures in the **Fubini–Tonelli theorem**.